



CGAN: closure-guided attention network for salient object detection

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Abstract

In recent years, salient object detection (SOD) has achieved significant progress with the help of convolution neural network (CNN). Most of the state-of-the-art methods segment the salient object by either aggregating the multilevel features from the CNN module or introducing the refinement module along with the baseline network. However, these models suffer from simplicity bias, where neural networks converge to global minima using the simple feature and remain invariant to complex predictive features. Very few methods concentrate on the neurophysiological behaviour of visual attention. As per Gestalt psychology, humans tend to perceive the objects as a whole rather than focus on the discrete elements of that object. The law of Closure (closed contour) is one of the Gestalt axioms that states that if there is a discontinuity in the object's contour, we perceive the object as continuous in a smooth pattern. This paper proposes a two-way learning network, where Closure-guided Attention Network (CGAN) and the Coarse Saliency Networks (CSN) jointly supervise the feature-channel to mitigate the simplicity bias. Furthermore, a channel-wise attention residual network is incorporated in the Closure Guided module to alleviate the scale-space problem and generate smooth object contour. Finally, the closure map from CGAN fused with the coarse saliency map of the Coarse Saliency Network generates a salient object. Experimental result on five benchmark datasets demonstrates the significant improvements in our approach over the state-of-the-art method.

Keywords Visual attention · Saliency · Encoder-decoder network · Gestalt · Convolution neural network · Multi-level fusion

1 Introduction

Visual attention is an active process, where cognition and perception play a significant role towards selective guidance of the gaze towards the most congruent object in the scene. The visually distinctive object in a particular scene is called the salient object. Intensive research has already been carried out in saliency detection over the last few decades. Most of the algorithms are based on bottom-up attention, where low-level features [1] such as colour, intensity, orientation, motion are used to determine the salient object [2, 3].

The bottom-up method is divided into several categories. In [4–6] the contrast priors have been considered for measuring the uniqueness of the colour feature to find the salient object. Some other works utilize the background

prior [7–10] to segment efficiently the salient object from the scene. Although contrast and background prior methods have helped achieve promising results, some limitations remain to identify the salient object from the image. For example, some non-salient areas of the image with a minuscule weight reduce the robustness of the saliency model. Thus some of the research works consider the object prior [11–14] method to successfully segment the salient object from an image by measuring the objectness [15]. However, all of the previous approaches successfully determined the saliency using characteristics of handcrafted features but failed in general. These models are not effective for all the real cases, particularly when the salient objects have a non-uniform appearance, complicated shapes, and the complex background, as shown in Fig. 1.

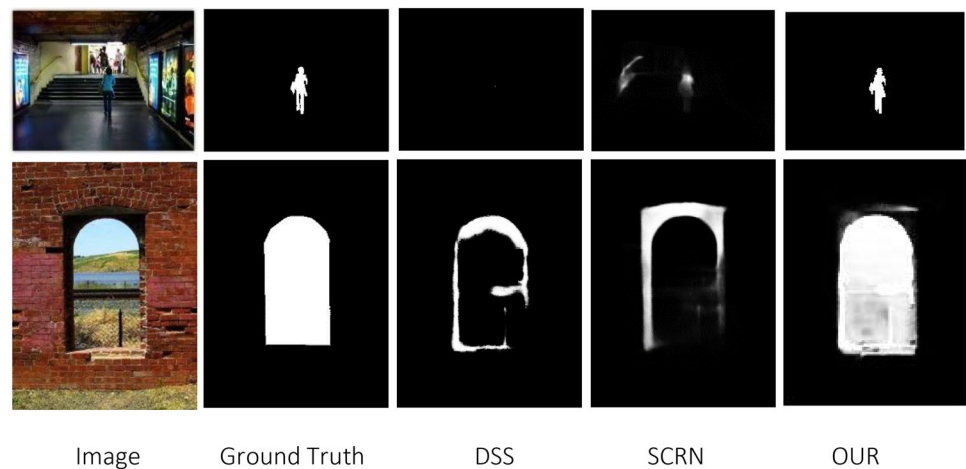
The recent advancements of Deep learning in computer vision have successfully overcome the above-mentioned problems with the help of CNN (i.e. VGG, ResNet). In the approaches [16–18], local features from the shallow network and global features from the deep network are aggregating hierarchically to generate a salient object. The most effective

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Fig. 1 Images with complicated shape and complex structure



SOD methods are currently based on Fully convolution networks (FCN). Considerable researches have been carried out by using short connection [19], side output [20], recurrent network [21] and refinement network [22]. However, all these networks are associated with two incomprehensible problems.

1. The deconvolution layer of an FCN may lead to scale-space problem and produce fuzzy saliency map in terms of contour and structural details.
2. Neural Network acts as a black box. It converges to global minima by learning the label data. However, we typically have little understanding of the decision through which it extracts the salient object. Such an obfuscated nature of the algorithm may lead to flawed bias (i.e. centre bias, human-centric object) and unable to find the saliency prior (i.e. objectness, closure).

Several psychological and neurophysiological studies on object-based attention reveal that attention and perceptual grouping are closely related to each other in biological visual systems [23]. According to Sensory Enhancement theory [23, 24], visual attention can spread along with the Gestalt cues, which are responsible for the perceptual grouping of the discrete elements into a meaningful object [25]. The closure is one of the primary Gestalt cues which strongly determine objectness of the particular image. Hess et al. [26], proposed that objects with closed and tight contours appear more prominent as a salient object than the object without closed contours. This significant features of visual attention termed as closure in several studies [27]. We introduce a two-way learning mechanism to overcome the problem mentioned above inspired by the above studies.

To overcome the first problem, we developed a Closure Guided Attention Network (CGAN), which generate closed contour (Closure) by using U-Net based [28] architecture. Most of the previous FCN methods failed to generate a precise contour map of the object due to the problem of

vanishing gradient in a deep network. We introduce residual operation on the selective channel of CGAN to overcome this problem. We can use ResNet as a baseline to resolve the above problem, but ResNet usually requires weeks for training, and thus it is practically infeasible for real-world applications. Moreover, the different channels of CNN have a response to different semantic information. The purpose of the salient object is to extract the most conspicuous information from the channel. Thus, it is unnecessary to focus on all the channels in the convolution layer to extract the salient object. Recently, attention modules have been employed in discriminative computer vision models to boost the performance of traditional CNNs. Instead of using all features parameter to extract the salient object, attention mechanism has been used in our case to minimize the number of parameter of the whole network. The problem of long range interactions has been tackled in sequence modelling through the use of attention layer. The ability of attention to learn to focus on important regions within a context has made it a critical component in neural transduction models for several modalities. Hence we apply channel-wise attention operation in the Closure guided network.

Thereafter, Coarse Saliency Network (CSN) has been incorporated to produce a coarse saliency map using the complementary information from the Closure-guided module. The problem of simplicity bias has been mitigated by exchanging information between two networks and simultaneously improving generalization. With the guidance of closure information, CSN concentrates on the objects those have the strongest contour and filters out the other non-salient features. VGG-16 [16] has been used as a backbone network to downsample the input image. We have added an extra dilation module to increase the filter's receptive field to fetch the high-resolution semantic information from the deeper network. Furthermore, decoder with Additive bilinear upsampling has been used to reconstruct the encoder

feature map into the saliency prediction map as the same size as the input image.

Finally, the closure feature map from CGAN has been fused with the saliency map of CSN to generate the final saliency map. Our contributions are summarized as follows:

1. We propose an encoder-decoder based Saliency Network to produce coarse saliency map, which is jointly supervising the feature channel along with the CGAN to eliminate the bias of the network.
2. We propose Closure-Guided Attention Network to generate an intermediate feature map that has the strongest closure prior. The residual attention network has been incorporated in the selective channel to successfully stitch the discrete contour into the closed contour or closure. We have used Dice Loss for CGAN and Cross-entropy Loss for CSN.
3. We propose a Fusion network, where an elementwise operation will fuse both the maps to produce an enhanced saliency map.
4. We have compared our model with ten state-of-the-art methods and achieved a better result under three evaluation metric on five widely used salient object datasets.

The remainder of the paper is organized as follows. Section 2 discusses the related works on salient object detection. Section 3 describes the proposed methodology of our network. Experimental results and performance evaluation are included in Sect. 4. Section 5 discusses the ablation studies. Sections 6 and 7 describe the runtime and limitation, respectively.

2 Related works

Saliency detection methods are divided into two categories. Early methods were based on the handcrafted features, which concentrated towards discovering the salient objects using contrast, background, colour, texture, and other visual cues. Current methods are based on the CNN based feature aggregation.

2.1 Handcrafted feature-based method

In the earliest work, [29] first described a computer implementation of cognitive modelling based on the primate visual system. Meur et al. [30–32] are proposed an approach for bottom-up saliency, which was based on the structure of the human visual system. Contrast sensitivity functions, perceptual decomposition, visual masking, and centre-surround interactions are some of the features implemented in this model. In [33], Shannon's self-information measure has been used for calculating the saliency of image regions. The

saliency of a local image region is the information that region conveys relatively to its surroundings. Li et al. [34] defined saliency using minimum uncertainty of a local region known as minimum conditional entropy [35, 36]. Liu et al. [37] proposed a set of novel features and adopted a Conditional Random Field to combine these features for salient object detection on their regional saliency dataset. Harel et al. [38] proposed Graph-Based Visual Saliency (GBVS) for salient object detection [39]. Duan et al. [40] proposed spatially weighted dissimilarity to measure saliency. Uniqueness and distribution of local features have been measured in [41]. Perazzi et al. [42] selected a specific component from an image and calculated selective contrast between adjacent clusters to distinguish the salient region from the non-salient region. Jiang et al. [43] improves regional-based contrast to define boundary extension and incorporates supervised learning-based smoothing operation to get the final saliency map. Shi et al. [44] investigated saliency by calculating the contrast in terms of the χ^2 distance between super-pixel regions to the background. In contrast to the previous method, [45] proposed a region contrast-based method to measure global contrast in the Laboratory colour space. Cheng et al. [46] developed the salient region by extracting the coarse data from bottom-up features and fused it with the top-down features extracted by the boundary-level propagation.

2.2 Traditional CNN method

While the existing handcrafted SOD approaches have demonstrated remarkable success in dealing with simple images, which usually consist of uniform objects and smooth backgrounds, they are likely to fail when the objects have nonuniform appearances or complex shapes, or when the backgrounds are vividly structured.

With the recent advancements of the convolution neural network, Salient Object Detection has achieved remarkable performance due to its capability of deep feature classification. Zhou et al. [47] used superpixel based CNN to extract the features for salient object detection. He et al. [48] proposed two deep neural networks to integrate local pixel estimation and global proposal search for salient object detection. Wang et al. [49] also predicts the degree of saliency of each superpixel by considering global and local context into account, and detects salient objects in a multi-context deep CNN. Zhao et al. [50] proposed a deep hierarchical saliency network to learn global structures and progressively refined the saliency map's details step by step via integrating local context information.

2.3 Fully convolution and refinement method

All of the above methodologies used specific deep features to model the saliency. But to get the more prominent results

in terms of resolution and global context, researches are boils down to the Aggregation based FCN method. Liu and Han [51] proposed an aggregating network where the multilevel feature maps integrate with the multiple resolutions to generate the final saliency map. Zhang et al. [19] introduced short connections to the skip-layer structures within the HED [52] architecture to find the saliency. Hou et al. [53] proposed leverage global and local pixel-wise contextual attention network to capture global and local context information. Liu et al. [20] proposed a bi-directional message-passing model to integrate multilevel features for salient object detection. All of the above FCN based Aggregation networks integrate with multiple features from different level to find the saliency. Sometimes, multilevel features contain redundant features that diminish the robustness of the model and produce a messy salient object. People used different refinement networks to produce finer output from coarse features to overcome this issue.

Zhang et al. [54] proposed a novel Edge-Aware network to refine the edge features and fuse them with the higher-level features to obtain the saliency. Wu et al. [22] introduced Boundary enhancement loss over the Attentive feedback network to mitigate the fuzzy boundary challenges to detect the salient object. Feng et al. [55] implemented a cross-feature module to fuse features from different level by an elementwise operation. They further used a cascaded feedback decoder to refine its high-level features from the previous stage to detect the saliency. Wei et al. [56] introduced a channel-wise attention module to assign a larger weight to specific feature channels responsible for saliency prediction.

Both the FCN-based Aggregation module and FCN-based Refinement module fail when multiobjects and complex structures appear in the image. None of the above methods concentrates on closure saliency prior, which is a primary factor in driving attention to the most conspicuous object. Wu et al. [22] assigned a larger weight to the boundary feature channel to get a more precise boundary. Nevertheless, the loss function cannot segregate the object with a closed boundary and an object with a sparse boundary. We have designed a guided loss function for the Closure network, which is updated simultaneously along with the saliency network. This bidirectional information exchange mechanism between the two-networks has successfully distinguished the object with a closed boundary from the object which does not have a closed boundary.

2.4 Attention/self-attention method

Attention mechanism [57] have become an integral part of effective sequence modelling and transduction models in various tasks, allowing modelling of dependencies without regard to their distance in the input or output sequences. Attention

has achieved higher rate of success in various computer vision tasks. Most notably, a channel-based attention mechanism termed Squeeze-Excite may be applied to modulate the scale of CNN channels [58] selectively. In their [58] works they incorporate the attention layer by explicitly modelling the interdependencies between the channel and its convolution features. The non-local [59] feature representation has been used to instantiate the self-attention network in video classification and object detection. A novel two-dimensional self-attention module has augmented into CNN to capture the long-range dependency for classification and object detection [60]. Bello et al. [61] worked on Convolution block attention module to focus on essential features and suppress the unnecessary one. Woo et al. and Chen and Li [62, 63] worked on attention mechanism as an auxiliary module along with the two-stream saliency network to extract the salient object by cross-modal interaction between the relevant channel. Inspired the above studies we implanted an attention module in encoder part to retain the long-term information relationship between subsequent feature maps. Our attention module helps to establish the contextual relationship between the feature maps which may elapse during the exhaustive training.

3 Methodology

Our proposed method has three modules. The first one is the Coarse Saliency Network, which generates coarse salient feature map. Second is the Closure-guidance Attention network, which generates closure prior map. Finally, Multilevel Feature Fusion Network has been implemented to generate the final saliency map. The overall architecture is illustrated in Fig. 2. Section 3.1 describes the underneath architecture of Encoder-Decoder module of Coarse Saliency Network. In Sect. 3.2, we have discussed the Closure Guided Attention Network and its residual learning capability. Section 3.3 describes the Fusion module's operation, where intermediate feature map forms CSN and CGAN are integrated by an elementwise operation to obtain the final saliency map. Section 3.4 describes the loss term of the overall network.

3.1 Coarse saliency network (CSN)

3.1.1 Encoder network

Our encoder model is based on VGG-16 architecture which was pre-trained on the ImageNet Dataset. We used all the convolutional layers of the VGG-16 except the FCN layer. Each convolution filter in these five blocks is restricted to a kernel size of 3×3 and operates with a stride of 1. All convolutional layers in the network are followed by Rectified Linear Unit (ReLU) activation to introduce elementwise nonlinearity. Following that, max-pooling with a kernel size

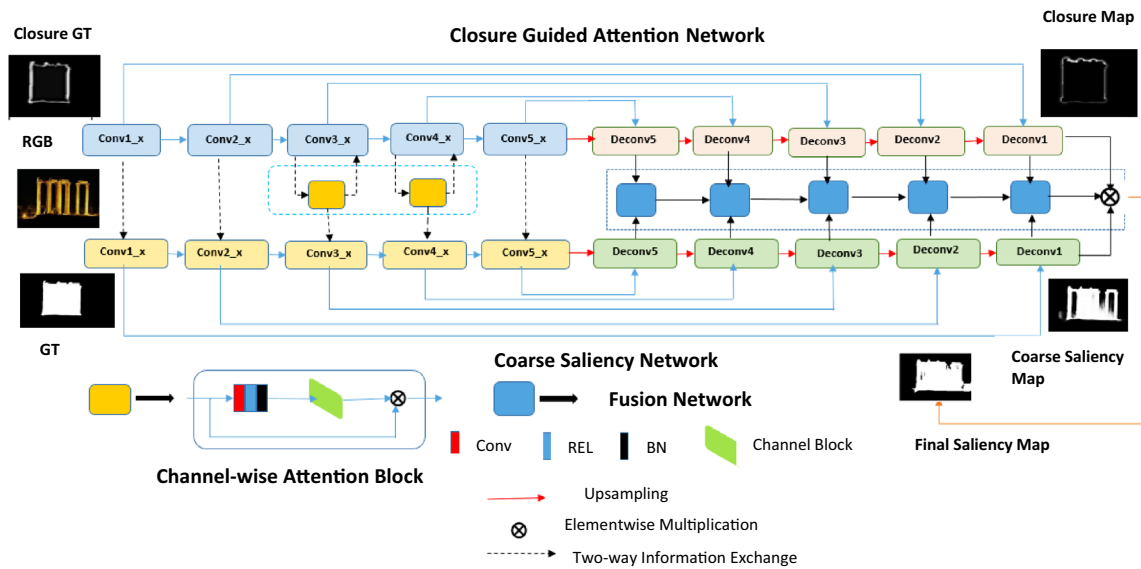


Fig. 2 Overall Architecture of the proposed model. Closure Guided Attention Network predicts closure map of the corresponding RGB image and Coarse Saliency Network predicts the coarse saliency map of the same RGB image. Channel-wise Attention Block (Yellow Box) has been incorporated to generate more adjoint and connected closure by residual learning, attention mechanism is further discussed in

2×2 and stride of 2 (non-overlapping window) is performed except on the Conv5 layer. Max-pooling is used to achieve translation invariance over small spatial shifts in the input image. The fourth and fifth block characterize the high-level semantic information while the initial convolution blocks extract low-level image cues such as colour, contrast and texture. Let us denote each convolution layer as E^i , where i is the number of the convolution layer in the encoder module. We obtain different feature map of an image from different convolution layers and can be defined as $F_{ENC}^{Coa} = (\sum_1^5 f_s(E^i), i = 1, 2, 3, 4, 5)$. Encoder module of CSN also takes the feature map as input from CGAN module and produces final feature map by minimizing the following objective function

$$F_{ENC}^{Coa} = \min \sum_i \mathbb{L}_{Coa}^{Clo}(f_s(E^i), f_c(E^i)) \quad (1)$$

where f_c denotes the feature parameter from the encoder module of CGAN and f_s denotes the feature parameter from the encoder module of CSN. $\mathbb{L}_{Coa}^{Clo}$ denotes the per-pixel loss function of encoder network with respect to closure network (CGAN).

The resolution of the features in the encoder network is heavily affected due to the appearance of the broader receptive field of pooling operations and convolution striding, which allows our network to learn the representation of

Sect. 3.2. Finally, Fusion block (Blue square box) has fused closure map and coarse saliency map to obtain the final saliency map. GT refers to the Ground truth of its corresponding image. Conv denotes as convolution block, Deconv denotes deconvolution block and BN stands for batch Normalization. Black dotted arrow indicates information exchange between two networks

abstract feature representation. To overcome this problem, we advocate using the atrous (Hole Algorithm) convolution, which utilizes several convolutional layers with different dilation rates in a parallel manner to capture multi-scale image information. We have used dilation layer after E_4 (con4_3) and E_5 (Conv5_3) with the rate of 4 and 8, respectively, to make the final extracted high-level features with scale and shape invariant.

No pooling layer has been used after Layer 5, so the spatial size of the final feature map is fixed at 28×28 , which is further upsample by the rate of 2 and pooling indices.

3.1.2 Decoder network

Decoder reverses the operation what encoder is doing. It upsamples the encoder image to their original size to preserve the spatial resolution. We use five decoder module as D^j , where j is the number of Decovolution layer in the decoder module. All the features from the previous decoder layer are upsampled by the rate of 2 using transposed convolution and obtain upsampled feature map which is defined as $F_{DEC}^{Coa} = (\sum_1^5 f(D)^j, j = 1, 2, 3, 4, 5)$, where F_{DEC}^{Coa} denotes the decoder module of Coarse saliency network and $f(D)^j$ is the feature parameter of each decoder module. In the decoder network, we performed transposed convolution by 5×5 learnable filter during upsampling. We use the BN

layer, and RELU Activation function in all the decoder layers same as encoder except the pooling layer. Instead of pooling layer, an un-pooling operation has been done on every subsequent decoder layer.

Moreover, all the pooling indices from encoder transfer to the subsequent decoder layer as side network to perform scale-invariant nonlinear upsampling of their input feature maps. After that, all the decoder and encoder channels are fused together. The softmax function has been adopted as an activation function to find the coarse saliency map with the size $H \times W$. Final Coarse saliency map is defined as

$$SAL_{Coa} = F_{ENC}^{Coa} + U(\text{Softmax}(\text{Trans}(F_{DEC}^{Coa}; \theta_S))) \quad (2)$$

where $U(\text{Trans}(*; \theta))$ denotes transposed convolution on the model parameter θ_S , let us consider an intermediate operation of Encoder-Decoder module with the feature parameter.

$$F_{DEC}^{Coa} = F_{ENC}^{Coa} \otimes F_{DEC}^{Coa} = \Phi^E(\omega_{x,y}, C^m) \otimes \Phi^D\{(\omega_{x,y}, C^m, \text{Trans}(f(D)^{j-1},))\} \quad (3)$$

Abbreviation for CSN: $f_S(E^i)$ = Feature parameter of each encoder module, $f_S(D^j)$ = Feature parameter of each decoder module, SAL_{Coa} = Saliency map obtain from CSN, F_{DEC}^{Coa} = Feature map of decoder module, F_{ENC}^{Coa} = Feature map of encoder module.

The final saliency map is generated by concatenating the encoder feature with the upsampled decoder feature. Where Φ^E , Φ^D denotes the RELU activation function of encoder and decoder module, respectively, $\omega_{x,y}$ is weighted feature parameter of x, y pixel of a particular Channel C^m .

3.2 Closure-guidance attention network (CGAN)

Our closure guided Attention network is trained on the Contour mask of the DUT-OMRON dataset and successfully estimated the closure prior with respect to the ground truth mask. We have incorporated the Contour-based ground Truth label for supervising CGAN, whereas the other models (DSS) used HED network to extract the finer boundary details. CGAN network designed upon a U-Net based architecture to accurately extract the sparse contour of the object. The U-net architecture is a fascinating combination of convolutions and transposed convolutions. Transposed convolutions work in a very similar way to convolutions, except that they use an upsampling kernel such that the spatial resolution is increased rather than decreased. Previous Fully Convolution Networks are suffering from the problem of localization accuracy due to their Max-pooling layers. Thus it is not easy to maintain good localization and context information at the same time. The U-net architecture eliminates the trade-off between localization accuracy and

the use of context by developing connections from the encoder side of the architecture. The loss function of the above operation is defined as:

$$L_{Clo} = 1 - \frac{\sum_{x=1}^H \sum_{y=1}^W SAL_{Clo}(x, y), GT(x, y)}{SAL_{Clo}(x, y) + GT(x, y)} \quad (4)$$

where $GT(x, y) \in \{0, 1\}$ is the ground-truth label of the contour map at pixel (x, y) and $SAL_{Clo}(x, y)$ is the predicted Closure map.

For a particular iteration in CGAN, a Closure map can be defined as

$$SAL_{Clo} = \text{cat}(F_{ENC}^{Clo}(x, y; \theta_C) \otimes F_{DEC}^{Clo}(x, y; \theta_C)); \\ i = 1, 2, 3, 4, 5; \quad j = 5, 4, 3, 2, 1 \} \quad (5)$$

where SAL_{Clo} denotes final closure map obtained from CGAN network, F_{ENC}^{Clo} is the feature map of the encoder module which is defined as $F_{ENC}^{Clo} = (\sum_{i=1}^5 f_C(E^i), i = 1, 2, 3, 4, 5)$, f_C denotes the feature parameter from the encoder module of CGAN. Feature parameter of each encoder module is transfer to the corresponding encoder module of the Coarse saliency network for learning the network together.

Cat() refers to the concatenation operation between encoder and decoder feature at pixel (x, y) . θ_C is the model parameter of the closure guided network. Closure map from each decoder module of CGAN has been transferred to the saliency network's decoder module for elementwise operation.

As the features pass through the Encoder, contour information gradually diminishes due to larger receptor of the filter and maxpooling operation. After completing the iteration, some of the low weighted contour areas have elapsed totally, and closure cannot be achieved. As per the Gestalt theorem, our brain fills any discontinuity presented in the contour pattern to perceive the pattern as a whole. Similarly, we assign a residual network to the Conv_3 and Conv_4 layer channel to generate aggregate closure features to learn the inter-channel relationship to mimic the Gestalt behaviour.

Accordingly, different feature channel has a response to different semantic information. All the channels are not responsible for keeping the saliency information. Thus, we developed channel-wise Attention in CGAN network to pass the relevant closure information to the subsequent layer. Recently, attention module has been incorporated in several computer vision model [58, 59] along with CNN to boost the performance of the overall module by mitigating the transduction problem.

Let us assume F_i be the input feature map coming out from the Conv4_x layer and F_{Attn} is the channel attention features. Then the elementwise operation is performed to get the refined closure feature F_{Refine} . Attention feature map and refine feature map can be denoted as follows.

$$F_{\text{Attn}} = \Phi^{\text{Attn}}(F_I^4(w_I^C, \theta)) \quad F_{\text{Refine}} = F_{\text{Attn}} \otimes F_I \quad (6)$$

C_{ij} is the 2D matrix, which represents the inter-channel relationship of the input feature map. Φ^{Attn} is the RELU operation on channel attention block, θ is the hyperparameter of the residual network. w_I^C is the weight vector of a particular channel which holds the salient information.

If $f_{x,y}$ is denoted as a feature of Channel C at pixel (x, y) , then updated closure feature map of 4th Encoder formulated in the spatial domain can be represented as

$$F_{\text{ENC}}^{(\text{Clo})}(f_{x,y}^4) = \Phi^E(f_{x,y}^4(w_{x,y}^C, \theta)) \oplus \sum_{k=1}^C F_{\text{Refine}} \quad (7)$$

where Φ^E denotes the RELU activation function of the encoder network. θ is the hyperparameter learn by the network. $w_{x,y}$ is weighted feature parameter at pixel location (x, y) (Fig. 3). *Abbreviation for CGAN module:* $f_C(E^i)$ = Feature parameter of each encoder module, $f_C(D^j)$ = Feature parameter of each decoder module, SAL_{Clo} = Saliency map obtain from CGAN, $F_{\text{DEC}}^{\text{Clo}}$ = Feature map of decoder module, $F_{\text{ENC}}^{\text{Clo}}$ = Feature map of encoder module

3.3 Fusion module

We design a Two-way fusion module to integrate both maps progressively by the elementwise operation to get a more refined salient object output.

Initially the decoder information from the CGAN is fused with the decoder module of saliency network by elementwise addition. As closure information is also complementary

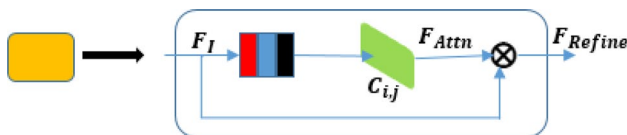


Fig. 3 Channel Attention Block with residual learning

to the salient object, saliency network can utilize more information before converging to global minima. After the fusion operation the final decoder information can be formulated as.

$$\text{SAL}_{\text{Fusion}} = [F_{\text{DEC}}^{\text{Coa}} \oplus F_{\text{DEC}}^{\text{Clo}}] \quad (8)$$

where $\text{SAL}_{\text{Fusion}}$ is the saliency information obtained from each of the fusion blocks after the decoder operation of two networks.

After obtaining the coarse saliency map and Closure map, the next is to integrate both the map in a single Saliency map by elementwise multiplication. Last fusion mechanism can be represented as

$$\text{SAL}_{\text{final}} = (\text{SAL}_{\text{Coa}} \otimes \text{SAL}_{\text{Clo}}) \otimes \text{SAL}_{\text{Fusion}} \quad (9)$$

where $\text{SAL}_{\text{final}}$ is the final saliency feature map (Fig. 4).

3.4 Supervision

In our proposed network, we use an RGB image and Binary ground truth for each image. Let us consider $I = \mathbb{R}^{H \times W \times 3}$ be the primary image, $G_s = \mathbb{R}^{H \times W}$ be the ground-truth saliency map and $\text{SAL}_{\text{final}} = \mathbb{R}^{H \times W}$ is the predicted saliency map. We have used Binary cross-entropy loss during the network training. The Loss term $\mathbb{L}_{\text{Coa}}$ defines as follows

$$\mathbb{L}_{\text{Coa}} = \sum_{x,y=1}^n G_s \log(\text{SAL}_{\text{Coa}}) + (1 - G_s) \log(1 - \text{SAL}_{\text{Coa}}) \quad (10)$$

But Cross entropy loss or BCE does not perform well in unbalanced labelling problems, like boundary or contour detection. Hence we use Dice loss in CGAN for its crisp boundary detection capability. Finally, the whole network was trained with a weighted sum of both losses. As mentioned earlier in Sect. 3.2, the saliency map can learn better by taking complementary information from Closure network.

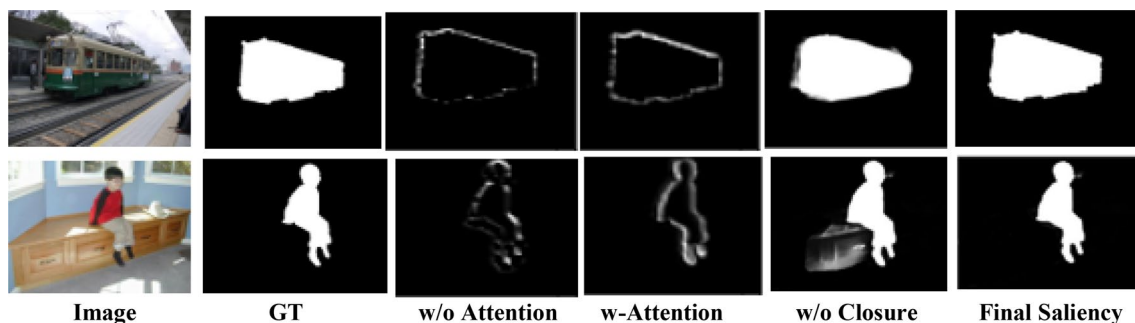


Fig. 4 Qualitative comparison of the proposed model with different component. 'w/o Attention' refers closure map without attention channel, w-Attention refers Closure with Attention Channel, w/o Clo-

sure denotes saliency map without the effect of CGAN and the last image represent the Final saliency map

The Loss term in CGAN is defined as

$$\mathbb{L}_{\text{Clo}} = 1 - \frac{\sum_{x=1}^H \sum_{y=1}^W \text{SAL}_{\text{Clo}}(x, y), GT(x, y)}{\text{SAL}_{\text{Clo}}(x, y) + GT(x, y)} \quad (11)$$

4 Experiment

4.1 Implementation

Our proposed method is implemented in Tensorflow API and training is carried out using Nvidia Jetson with 16 GB of RAM and 4 GB Denver GPU. The DUTS-OMRON dataset are used to train the saliency network and CGAN. ContourGAN [64] algorithm has been used to obtain the cloure-map ground truth to train the CGAN. We used VGG-16 as the backbone network, in which the parameter is pretrained on ImageNet dataset. We set the learning rate as 0.0003 for both the network. Both networks have learned by using ADAM optimizer, and batch size was set to 12. During the training, all the layers are initialized by ‘He’ weight initializer due to its stability on RELU activation function. We used a pretrained VGG-16 model from the Tensorflow model Zoo to train our CSN network on pretrained VGG-16 where final layer of softmax function has been cut down to two from 1000. Similarly, U-Net model has been used in the case of CGAN. Unlike VGG-16, UNet model was not pretrained on ImageNet datasets and we used it from scratch by rewriting its encoder part. The attention part of our proposed method has been incorporated in the encoder part of the UNet model. Finally the CSN module trained on DUT-OMRON datasets and its takes 8 h to converge on global minima with 60 epoch. Similarly, CGAN module also trained on DUT-OMRON with its closure ground truth and takes 6 h to converge on global minima with 55 epochs. The inference time for an image with size 2243224 is 0.021 s.

4.2 Datasets

For the evaluation of our proposed algorithm, we have performed experiments on five publicly available datasets: ECSSD, PASCAL-S, DUT-OMRON, HKU-IS, DUTS-TE. ECSSD has 1000 images with multiple objects and structurally complex scenes. HKU-IS has 4447 images, which includes multiple disconnected objects or objects touching the image boundary. DUT-OMRON contains 5168 high-quality images with a relatively complex background and multiple object appearance. DUTS-TE is the largest salient object detection benchmark dataset. It contains 10,553 images for training and 5019 images for testing. PASCAL-S dataset contains 850 natural images from the PASCAL VOC dataset.

4.3 Evaluation metrics

We evaluate the performance using three metrics: P-R curve, Mean Absolute error and Max F-measure. Precision is defined as the percentage of correctly segmented salient object pixels to all pixels of the segmented salient object and recall are defined as the percentage of correctly segmented salient object pixels to all pixels of the ground truth. The precision and recall are defined as

$$\text{Precision} = \frac{\text{SAL}_{\text{final}} \cap G_S}{\text{SAL}_{\text{final}}}$$

$$\text{Recall} = \frac{\text{SAL}_{\text{final}} \cap G_S}{G_S}$$

where $\text{SAL}_{\text{final}}$ is the Binary mask of the saliency map and G_S is the Ground truth binary map.

The F measure (F1 score or F score) is a measure of accuracy of the test and is defined as

$$F\beta = \frac{(1 + \beta^2) \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \quad (12)$$

We have used $\beta^2 = 0.3$ to put a higher weightage on precision than recall.

MAE refers to the average pixel-wise error between the saliency map and the ground truth. MAE is the average of the absolute difference between the predicted values and observed value. The MAE is a linear score which means that all the individual differences are weighted equally in the average. It measures the average magnitude of the errors in a set of predictions, without considering their direction. It is defined as

$$\text{MAE} = \frac{1}{WXH} \sum_{i=1}^W \sum_{j=1}^H |S(i, j) - G(i, j)| \quad (13)$$

where is $S(i, j)$ predicted saliency map and $G(i, j)$ is the corresponding ground truth map.

S-measures are used to evaluate the structural similarity of the predicted non-binary saliency map and the ground truth.

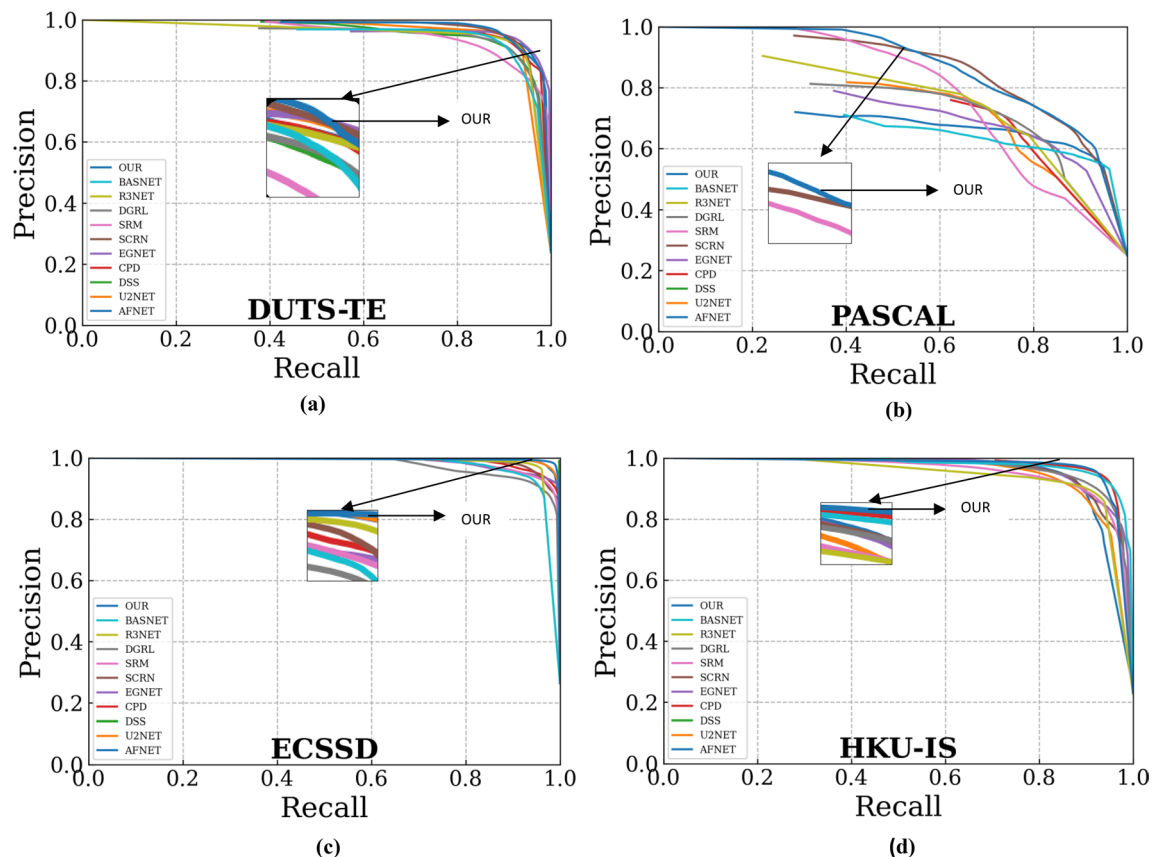
4.4 Comparison with other methods

We compare our model with ten state-of-the-art models including.

DSS [19], SRM [18], R3Net [21], SCRNet [54], EGNet [65], DGRL [66], CPD [17], AFNet [49] BASNET [67], U2NET [68]. For comparing the results are either re-produced using the author's source code or referred from their Google drive. We evaluate and compare with other models in terms of maxF, Structure Measure and MAE, as shown in Table 1.

Table 1 Comparison of our model with ten state-of-the-art models on ECSSD, HKU-IS, PASCAL-S, DUTS-OMRON, DUTS-TE in terms of MaxF($\uparrow$) S($\uparrow$) MAE($\downarrow$). Red and green indicate the best and second-best performance

Model	Dataset														
	ECSSD			HKU-IS			PASCAL-S			DUT-OMRON			DUTS-TE		
	MaxF	S	MAE	MaxF	S	MAE	MaxF	S	MAE	MaxF	S	MAE	MaxF	S	MAE
AFNet	0.935	0.914	0.042	0.923	0.905	.036	0.868	0.850	0.071	0.797	0.826	0.057	0.862	0.866	0.046
CPD	0.939	0.918	0.037	0.925	0.906	.034	0.861	0.848	0.071	0.797	0.823	0.056	0.865	0.869	0.043
DGRL	0.921	0.906	0.043	0.910	0.896	.036	0.844	0.839	0.075	0.774	0.791	0.062	0.828	0.836	0.049
EGNET	0.943	0.918	0.041	0.937	0.918	.031	0.869	0.852	0.074	0.842	0.818	0.052	0.893	0.875	0.039
SCRN	0.950	0.927	0.038	0.934	0.917	.034	0.882	—	0.064	0.812	0.837	0.056	0.888	0.885	0.040
R3Net	0.934	0.910	0.040	0.915	0.895	.036	0.834	0.807	0.092	0.795	0.817	.063	0.828	0.817	0.058
SRM	0.916	0.895	0.056	0.906	0.887	0.046	0.838	0.832	0.084	0.769	0.777	0.069	0.826	0.824	0.058
DSS	0.916	0.882	0.052	0.910	0.879	0.041	0.831	0.798	0.093	0.771	0.788	0.066	0.825	0.822	0.057
BASNET	0.942	0.916	0.037	0.928	0.909	0.032	0.856	0.838	0.076	0.805	0.836	0.056	0.860	0.853	0.047
U2NET	0.951	0.928	0.033	0.935	0.915	0.032	0.859	0.844	0.074	0.823	0.836	0.054	0.860	0.853	0.047
Our	0.986	0.978	0.021	0.987	0.957	0.033	0.812	0.801	0.086	0.898	0.856	0.055	0.912	0.899	0.032

**Fig. 5** Quantitative results with PR-curve on four widely used benchmarks: **a** DUTS-TE, **b** PASCAL, **c** ECSSD, **d** HKU-IS

As shown in Fig. 5 our method surpasses other methods on all the dataset in terms of P-R curves. EGNet and SCRNet also provide a better result on those datasets. But in terms of F-measure, our model stands in a better place

than any other method on DUTS-OMRON, DUTS-TE and HKU-IS dataset. These datasets contain images with spurious contour information, which can only be recovered by estimate the closure prior in a successive manner. Table 1

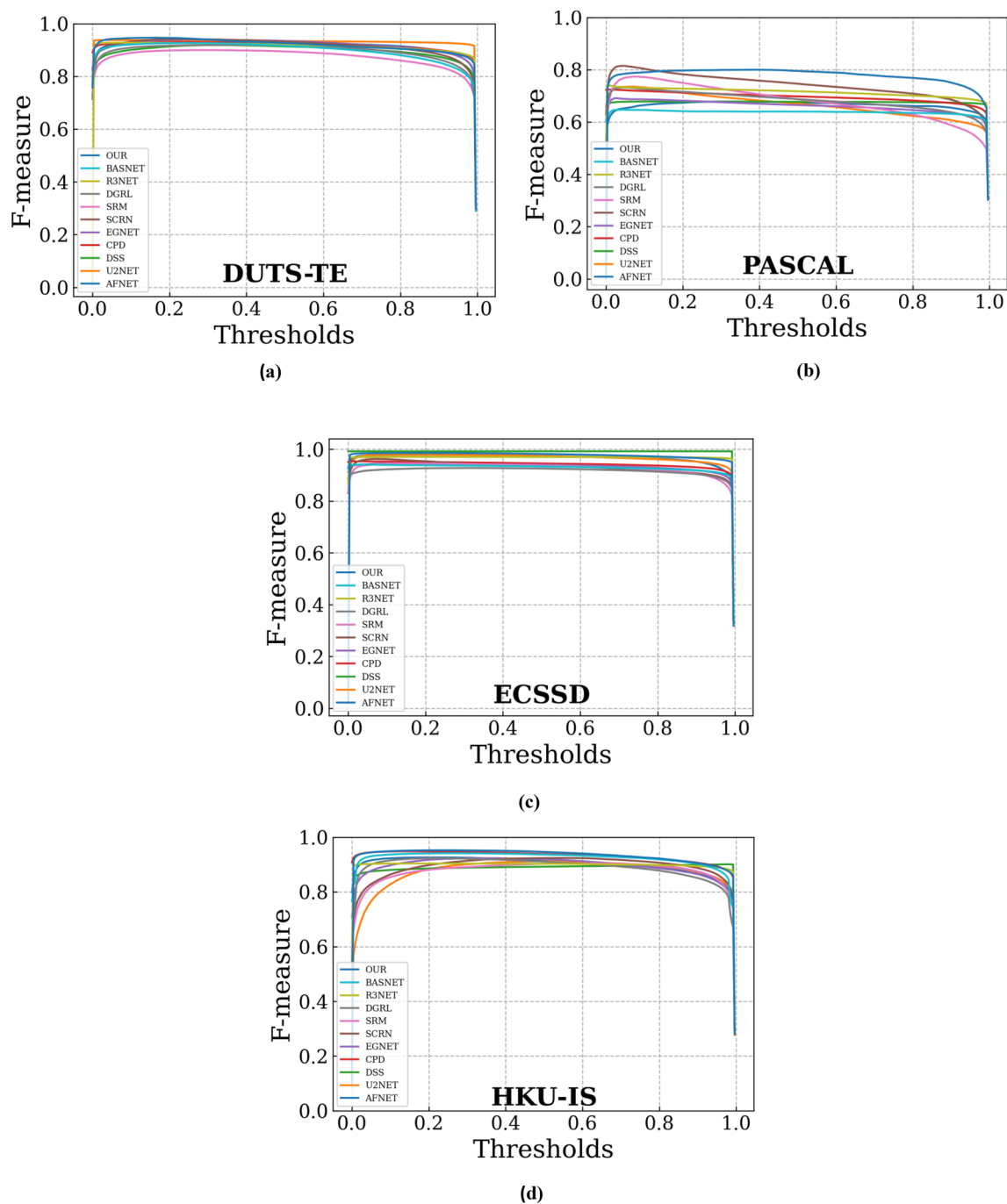


Fig. 6 Quantitative results with F-measure on four widely used benchmarks: **a** DUTS-TE, **b** PASCAL, **c** ECSSD, **d** HKU-IS

shows the comparative result in terms of maxF, S, and MAE. Our method exhibits the best performance in ECSSD, DUT-OMRON and DUTS-TE dataset in terms of all the parameters but failed in the case of PASCAL dataset. We used AFNet, EGNET and SCRNet, where the boundary and edge features were considered to evaluate the saliency for a fair comparison. The above-mentioned models failed to achieve better results in the DUT-OMRON and DUTS-TE dataset

due to the presence of intricate and multiobject in the image as represented in Fig. 1.

4.4.1 Quantitative comparison

The Precision-Recal and F-measure curve are shown in Figs. 5 and 6, respectively.

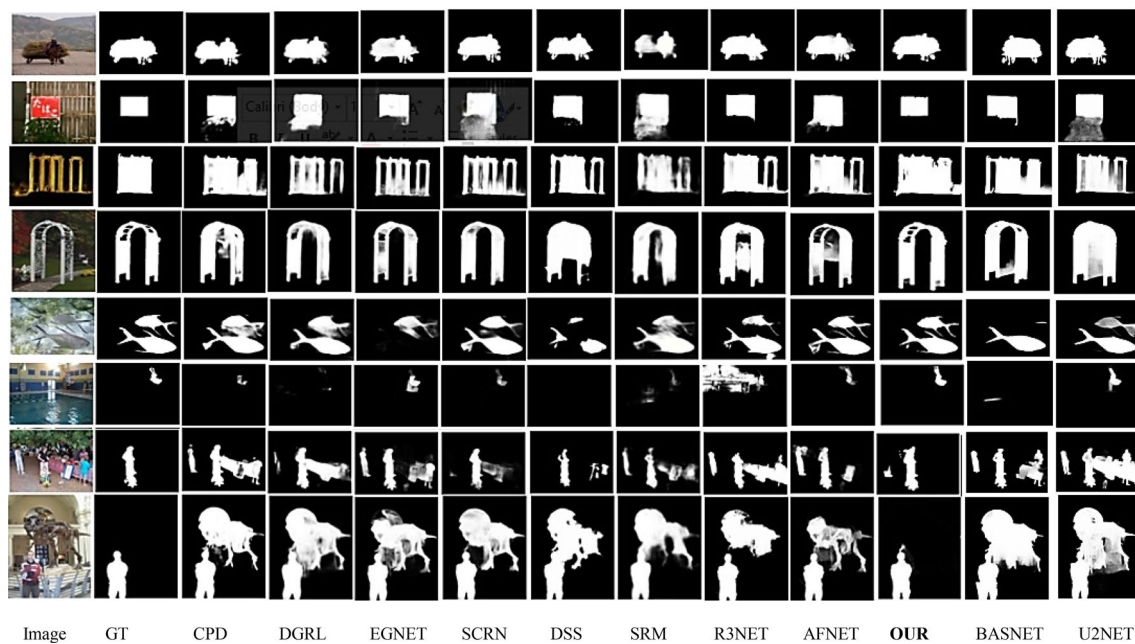


Fig. 7 Qualitative comparison of the proposed model with ten state-of-the-art methods

4.4.2 Qualitative performance

Qualitative results of our model and ten state-of-the-art models are presented in Fig. 7. The first and second rows present a single object in the simple scene, third and fourth row presents a single object with a complex structure. Similarly, fifth and sixth rows present low contrast image where saliency between the background and foreground is diminished. The most vital and complicated images are represented in the rows 7 and 8, where the images contain multiple objects in a complex environment. Most of the methods failed drastically to detect the salient object in the above-mentioned situation. As shown in the last row, all the methods detect two objects instead of a single salient object. But in our method, the salient object successfully pops out from the image due to channel-wise attention. Similarly, in row 6, EGNET [65] and AFNet [22] successfully extracted the salient object but cannot highlight the leg's boundary. This problem can also be solved by the use of DICE loss in CGAN (Sect. 3.3). Our proposed method can produce a better result closer to the ground truth than the other state-of-the-art methods as a whole.

5 Ablation study

We discuss the contribution of each module in this section. To verify each module's effectiveness, we evaluate CGAN, CGAN + CSN and CGAN + CSN + Fusion separately on five datasets in terms of maxF, MAE and S measure. Qualitative and Quantitative results are presented in Fig. 8 and Table 2, respectively. Each module has a significant role in improving the accuracy of the saliency map. Coarse saliency network predicts the coarse saliency map by learning the ground truth data. But this module was unable to achieve the higher accuracy without the help of closure cue. Thus CGAN Module and its channel-wise residual operation are incorporated to restrict the network from vanishing gradient problem. Residual operation of this module helps to overcome the contour degradation problem. Figure 8 shows the visual results obtain from each module, Fig. 9 depicts the visual results of the closure map with different weight vectors for each iteration.

As shown in Fig. 9, the weight parameters are gradually decreased during the backward propagation, and the object contour is diminished at the end of the iteration. Thus we

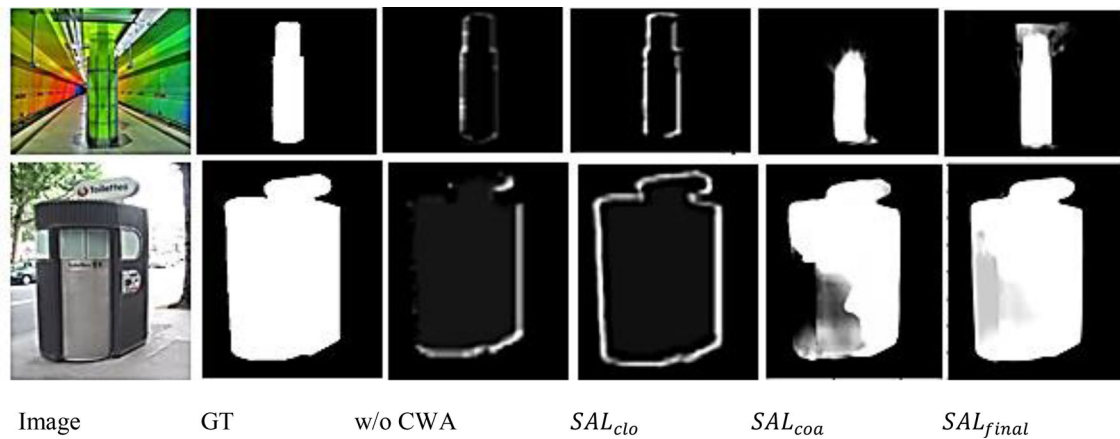


Fig. 8 Qualitative comparison of our model with different component. ‘w/o CWA’ refers closure map without attention channel, SAL_{clo} represent the Closure with Attention Channel, $xxxxx$ denotes coarse

saliency map without the effect of CGAN and the SAL_{final} represents the Final saliency map after fusion

Table 2 Comparison of our model with and without CGAN on ECSSD, HKU-IS, PASCAL-S, DUTS-OMRON, DUTS-TE in terms of MaxF($\uparrow$) MAE($\downarrow$). Red and green indicate the best and second-best performance

Model	Dataset									
	ECSSD		HKU-IS		PASCAL-S		DUT-OMRON		DUTS-TE	
	MaxF	MAE	MaxF	MAE	MaxF	MAE	MaxF	MAE	MaxF	MAE
Only CSN	0.835	0.105	0.809	0.145	0.782	0.152	0.792	0.123	0.789	0.154
CSN + CGAN	0.945	0.024	0.902	0.068	0.803	0.086	0.892	0.068	0.856	0.083
Our(after Fusion)	0.986	0.021	0.987	0.033	0.812	0.086	0.898	0.055	0.912	0.032

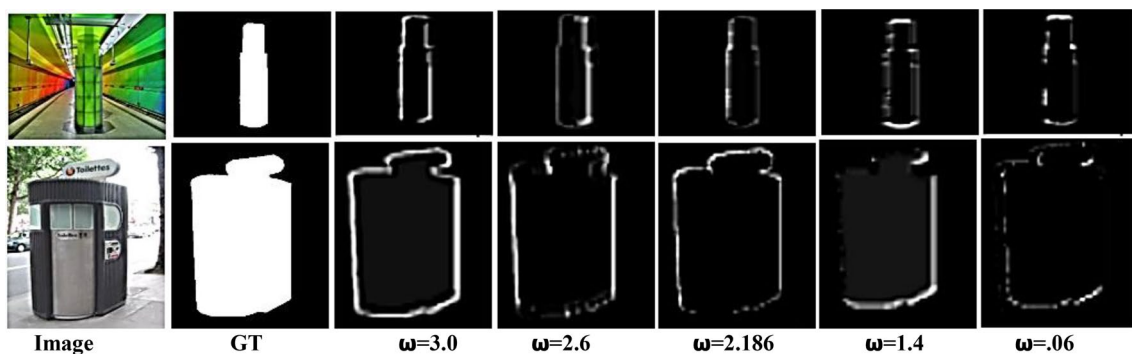


Fig. 9 Qualitative comparison CGAN module without the effect of channel-wise residual operation in successive iteration

introduce residual operation on the selective channel to retain the object contour intact and unbroken. We also conduct a quantitative experiment on the five datasets to demonstrate the effectiveness of the different modules in terms of maxF and MAE.

From the above observations, we can conclude that (1) coarse saliency alone cannot produce a desirable result. The mean absolute error is always higher in the CSN network for all five datasets. (2) After incorporating the CGAN module, the mean absolute error decreased by approximately 7% (0.3). As shown in Fig. 8, the final

saliency map (SAL_{final}) predicts a more accurate result than CSN with the help of the CGAN module.

6 Runtime comparison

We compare the inference time of our method with the others using the same GPU. Our network takes 0.076 s to process an input image of size 300*400. Table 3 represents the comparison of our method with other state-of-the-art method in terms of inference time. As shown in Table 3,

Table 3 Approximate runtime of different methods with their baseline model

Model	CPD	DGRL	EGNET	SCRN	DSS	SRM	R3Net	AFNet	OUR
Time (s)	0.048	0.16	0.035	0.019	0.043	0.027	0.034	0.040	0.021
Baseline	VGG-16	Inception	VGGNet	U-NET	VGGNet	Resnet-50	ResNet	VGG-16	VGG-16+ U-Net

we achieve the second position in terms of inference time. Since our method aims to improve the accuracy, the runtime of our approach does not achieve remarkable performance.

7 Limitation and future works

It is very much essential to point out the limitations of our model for further improvements. Figure 10 displays the results of the failure cases. For instance, our method fails (i) when an object had a minimal size (top row of Fig. 10), it seems that a small object's feature may be elapsed during the exhaustive iteration (ii) when a single object appears as disconnected one by occlusion (middle row) (iii) for multiple objects with similar contrast (last row). Our CGAN module cannot identify the object's closed contour for both of the images due to contrast variability and occlusion. As future study, we plan to incorporate the multimodal fusion technique with a more challenging scenario, such as a small object with an occluded boundary and an object with variable contrast. Moreover, a transformer-based vision strategy may be incorporated efficiently along with the network to mitigate long-distance interaction problems.



Fig. 10 Failure examples. **a** Input images, **b** ground truth, and **c** predicted result of the proposed method

8 Conclusion

This paper proposes a novel saliency detection model to alleviate simplicity bias and segment the salient object in terms of the closed contour or closure. The closure guided network guides the coarse saliency network to focus on the object with the strongest closure. A channel-wise attention module is further utilized to select the relevant feature channel and negate the other redundant channel. CNN with the deeper network cannot produce a fine-grained contour due to its small receptive field and max-pooling operation. The residual operation has also been incorporated into the selective channel to refine the output closure map to mitigate the above issue. A hybrid loss function has been used to preserve saliency information in both the network. Finally, the output of the closure network is fused with the coarse saliency network to get the salient object by the elementwise operation. The experimental result on the five benchmark dataset demonstrates that our model achieves better performance than others.

Declarations

Conflict of interest We declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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